

Baiying Lu

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OVERVIEW

Ph.D. candidate in Computer Science specializing in **applied machine learning, time-series modeling and mobile computing**, with industry experience in **LLM fine-tuning, foundation model, and large-scale recommendation systems**. Actively seeking a **Full-time ML/AI Engineer, Software Engineer, or Research Scientist** role.

EDUCATION

Dartmouth College <i>Ph.D. in Computer Science</i>	Hanover, NH 2021 – Present (Expected Feb 2027)
Duke University <i>M.S. in Biomedical Engineering; GPA: 3.97/4.00</i>	Durham, NC 2019 – 2021
Shandong University <i>B.E. in Biomedical Engineering; GPA: 91.13/100</i>	Shandong, China 2015 – 2019

INTERNSHIP EXPERIENCE

ByteDance – Commerce Ads Technology

Machine Learning Scientist Intern

San Jose, CA Jun.-Sep. 2026

- Developed an **LLM-based generative retrieval** branch for large-scale recommendation based on hierarchical Semantic IDs (SIDs) on a candidate space of **thousands million items**.
- Built and scaled **full-parameter LLM supervised fine-tuning** pipelines on **millions of user behavior sequences**, systematically optimizing prompt construction, model scale, training objectives, and semantic grounding for generative recommendation.
- Designed **trie-constrained beam search** for valid SID generation, achieving **~5.9% Recall@1** and **~29.9% Recall@50**; improved output diversity and Top-K retrieval quality while mitigating output collapse and over-concentration.

Samsung Research America – Digital Health Team

Research and Development Intern

Mountain View, CA Jun.-Sep. 2025

- Designed and implemented a scalable end-to-end data pipeline to ingest, preprocess, and manage 1M+ real-world smartwatch PPG segments, enabling large-scale model training and evaluation.
- Built and optimized deep learning models (ResNet, VGG, attention-based architectures) for arrhythmia detection, achieving >85% F1 while reducing per-segment inference error to <0.05 MAE.
- Integrated an arrhythmia detection task into a multimodal healthcare foundation model, contributing reusable components to an internal ML platform used by multiple downstream teams.

Novo Nordisk – AI Analytics, Digital Data Science Dept.

Data Scientist Intern

Boston, MA Jun.-Aug. 2024

- Built reusable data analysis pipelines in Python to process and analyze large-scale clinical wearable datasets (300+ patients, 30+ days per subject).
- Implemented feature extraction and aggregation workflows for continuous glucose monitoring and activity data, supporting downstream ML modeling.
- Developed statistical and ML-based analyses to identify key behavioral and physiological factors affecting glucose control, generating insights adopted by digital therapeutics teams.

RESEARCH EXPERIENCE

Enabling Data-driven Health Management with Mobile Technology

Dartmouth College

Student Researcher, HealthX Lab

Sept. 2021 – Present

- Developed cross-platform mobile clients supporting both Android (Kotlin) and iOS (Flutter), enabling secure data synchronization from Dexcom CGM APIs and Fitbit APIs for continuous wearable data collection.
- Built and deployed backend services on Google Cloud Platform using Go, with Firebase for user management and scalable storage of high-frequency time-series sensor data.
- Lead a benchmarking project evaluating state-of-the-art time-series foundation models, LLM for time series, and deep learning architectures, including Chronos, TimesFM, TimeLLM, LSTM, and Transformers for blood glucose prediction using uni- and multimodal data from medical- and consumer-grade wearables.
- Built and maintained end-to-end research data pipelines covering study design, data collection, preprocessing, feature engineering, and statistical analysis for multiple digital health studies.

Digital Biomarkers Processing Pipeline from Wearable Devices

Duke University

Research Assistant, Big Ideas Lab

Jan. 2020 – June 2021

- Developed a data compression and transformation pipeline for digital biomarkers from wearable sensors, implementing data processing and transformation method, such as Wavelet Transforms, Autoencoders, and Huffman Encoding, to achieve up to $130\times$ compression with under 2% information loss (published in *Sensors*, 2021).
- Collaborated with clinicians and behavioral scientists on an IRB-approved study assessing patient perspectives on wearable data sharing, contributing to study and survey design and manuscript preparation (published in *npj Digital Medicine*, 2021).

Depression Level Classification Using Three-Channel EEG Data

Shandong University

Undergraduate Research Assistant, Licai Yang Lab

Jan. 2019 – June 2019

- Implemented a EEG preprocessing pipeline using band-pass Butterworth filtering, Hilbert–Huang Transform, and Empirical Mode Decomposition to extract α -band activity.
- Engineered time- and frequency-domain features and applied SVM and Random Forest models to classify depression levels using clinical three-channel EEG recordings from real-world patients.

SELECTED PUBLICATION

B. Lu, ..., C. Kang, “Feasibility of Ectopic Beat Detection and Count Estimation from Smartwatch-Based Photoplethysmography,” Accepted by ICASSP, 2026.

B. Lu, ..., T. Prioleau, “Deep learning for blood glucose prediction: reproducibility challenges and factors affecting differential performance,” Accepted by PLOS Digital Health, 2026.

B. Lu, ..., S Tang, T. Prioleau, “GlucoFM-Bench: Benchmarking Time-Series Foundation Models for Blood Glucose Forecasting,” Under Review, NeurIPS, 2026.

B. Lu, ..., X. Zhou, G. Forlenza, T. Prioleau, “Mealtime Prediction using Wearable Insulin PumpData to Support Diabetes Management,” Scientific Reports - Nature, 2024.

P. Belsare, **B. Lu**, A. Bartolome, T. Prioleau, “Investigating Temporal Patterns of Glycemic Control around Holidays,” IEEE Eng. in Medicine and Biology Conference, July 2022.

B. Bent, **B. Lu**, J. Kim, J.P. Dunn. “Biosignal Compression Toolbox for Digital Biomarker Discovery”. *Sensors*, 2021.

SKILLS

Programming: Python, Kotlin (Android), Go, Flutter/Dart, React Native, C, C++, Java, JavaScript, SQL, Bash, Git, MATLAB, Linux.

Machine Learning & Data: PyTorch, TensorFlow, LLM post-training, LLM Fine-tuning, Generative recommendation, Retrieval & Ranking, Scikit-Learn, SciPy, LSTM, CNN, Transformer, time-series foundation model, data preprocessing & analysis, biosignal preprocessing&modeling, hyperparameter tuning, model evaluation & reproducibility

Cloud & Tools: AWS (S3, SageMaker), Distributed computing, vLLM, SGLang, GCP, Firebase (Firestore, Cloud Functions), HuggingFace, Docker, REST APIs, real-time data streaming, model online serving.

Experience Areas: Deep Learning, Machine Learning, Recommendation System, Statistical Analysis, Artificial Intelligence, Algorithms, Data Science, Mobile App Development, Human–Computer Interaction.